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Revolutionizing Drilling Fluids: Granite Waste as a Cost-Effective Bridging Material

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Abstract

The ongoing demand for sustainable and economically viable drilling fluid additives is reshaping material selection in the oil and gas sector. This study investigates the potential of repurposing industrial granite waste as a bridging material in water-based drilling fluids, offering a cost-effective and environmentally responsible alternative to conventional barite. Four distinct mud formulations were systematically evaluated: baseline control without bridging material, a conventional fluid with 100% barite, a hybrid blend with 50% barite and 50% granite waste by weight, and a formulation employing 100% granite waste as the sole bridging agent. The granite waste was extensively characterized using particle size distribution analysis, X-ray fluorescence (XRF), X-ray diffraction (XRD), and scanning electron microscopy (SEM), confirming its suitable mineralogy, chemical composition, and particle structure for drilling fluid applications. Laboratory measurements assessed fluid density, pH, and viscosity at 120°F, while filtration performance was initially screened using API tests at room temperature and 100 psi. The most promising formulations underwent rigorous high-temperature, high-pressure (HTHP) filtration tests at 230°F and 300 psi with a 20 µm ceramic disk to simulate challenging downhole conditions. The results revealed that the drilling fluid containing 100% granite waste achieved outstanding fluid loss control, reducing filtrate volume to only 5 mL after 30 minutes outperforming both the traditional barite and hybrid systems. These findings demonstrate that granite waste is not only a technically effective bridging material but also a sustainable solution that promotes resource circularity and reduces environmental impact in drilling operations. The research paves the way for future optimization of waste-derived additives in advanced drilling fluid design.

Introduction

The drilling industry is increasingly challenged to develop cost-effective and environmentally sustainable drilling fluid formulations, as both regulatory requirements and stakeholder expectations demand greater operational responsibility. Fluid loss control remains a persistent technical hurdle, directly affecting wellbore integrity, operational efficiency, and environmental outcomes. Traditionally, barite has been employed as the bridging agent of choice in water-based drilling fluids due to its proven effectiveness in

sealing permeable formations. However, barite use is associated with several drawbacks, including high and volatile costs, reliance on imports, and the environmental burden of mining, processing, and long-distance transportation (Mahmoud, Husameldin, et al., 2024).

In response to these challenges, the industry is increasingly focused on the valorization of industrial byproducts and waste streams as alternative functional additives in drilling operations (Chernova et al., 2021). Granite waste, a readily available byproduct of the dimension stone industry, offers compelling potential as a sustainable bridging material. Notably, in Saudi Arabia alone, annual granite waste production was estimated at approximately 220,000 to 275,000 metric tons as of 2018, reflecting both the scale of the issue and the substantial resource available for beneficial reuse. Characterized by its chemical inertness, suitable particle size distribution, and mechanical robustness, granite waste is particularly well-suited for repurposing as a drilling fluid additive. Utilizing granite waste not only diverts significant volumes of industrial byproduct from landfills but also reduces the sector's dependence on conventional mined materials, contributing to the broader goals of resource circularity and environmental stewardship.

This study presents a systematic investigation into the feasibility of employing granite waste as a bridging agent in water-based drilling fluids, directly comparing its performance to that of barite. Four distinct mud formulations were prepared for evaluation: (1) a baseline control sample without bridging material, (2) a conventional formulation containing 100% barite, (3) a hybrid formulation with 50% barite and 50% granite waste by weight, and (4) a formulation with 100% granite waste as the sole bridging agent. All formulations were engineered to maintain comparable rheological properties, with particle size distributions of barite and granite waste closely matched to ensure fair comparison.

Comprehensive laboratory characterization of granite waste included particle size distribution analysis, X-ray fluorescence (XRF), X-ray diffraction (XRD), and scanning electron microscopy (SEM) to assess its mineralogical and physical suitability for use in drilling fluids. The performance of each formulation was evaluated through a series of laboratory tests, with particular emphasis on High-Temperature High-Pressure (HTHP) fluid loss testing at 230°F and 300 psi to simulate demanding downhole conditions. Fluid loss after 30 minutes was recorded as the key performance indicator. The results demonstrate that the drilling fluid containing 100% granite waste as the bridging agent achieved an exceptionally low fluid loss of 5 mL, outperforming both the baseline and conventional barite-based formulations.

By rigorously demonstrating the technical viability of granite waste as a replacement for barite, this research advances the application of industrial waste materials in drilling operations. The findings highlight granite waste's potential to deliver cost-effective, sustainable, and high-performance fluid loss control, while simultaneously reducing the environmental impact of both drilling fluid formulation and industrial waste disposal. These insights contribute to the growing body of literature supporting the integration of circular economic principles into oil and gas operations and underscore the importance of continued research into the optimization of waste-derived additives for advanced drilling fluid systems (Ali et al., 2023; Mahmoud & Elkatatny, 2020).

Materials and Method



Figure 1—Methodology workflow.

Granite Waste Processing

Granite wastes were ground and sieved to obtain a fine powder for use as a potential fluid loss additive in drilling fluids.

Material Characterization

The particle size distribution (PSD) analysis Fig 2 showed that the powder ranged from 45 μm to 300 μm , with a median particle size (D50) between 100–150 μm , depending on the milling conditions. Finer particles enhance dispersion and improve fluid loss control in drilling muds. The particle size was measured using laser diffraction and verified through sieve analysis.

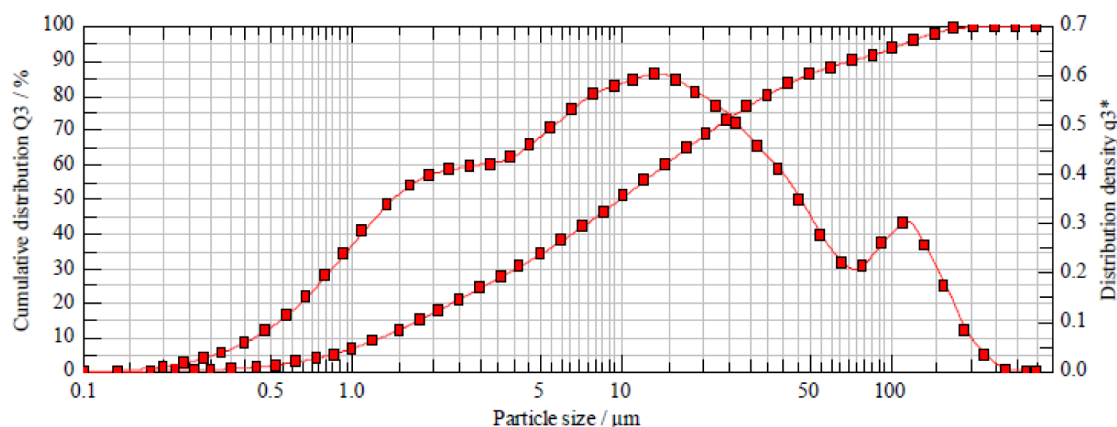


Figure 2—Particle size distribution of granite waste powder.

The SEM-EDX analysis Fig 3 shows the elemental composition and morphology of the examined sample. At a magnification of 750x, the microstructure reveals a heterogeneous surface. The elemental composition, determined by Energy Dispersive Spectroscopy (EDS), primarily consists of oxygen (O) at 46.02 wt%, silicon (Si) at 32.33 wt%, iron (Fe) at 4.80 wt%, potassium (K) at 4.44 wt%, aluminum (Al) at 7.07 wt%, sodium (Na) at 3.41 wt%, and calcium (Ca) at 1.93 wt%. The high silicon and oxygen content indicates the dominance of silicate minerals, while the presence of iron and aluminum suggests typical components

found in clay or feldspar minerals. The spectrum further confirms the mineralogical characteristics relevant to drilling mud additives or geological analysis applications.

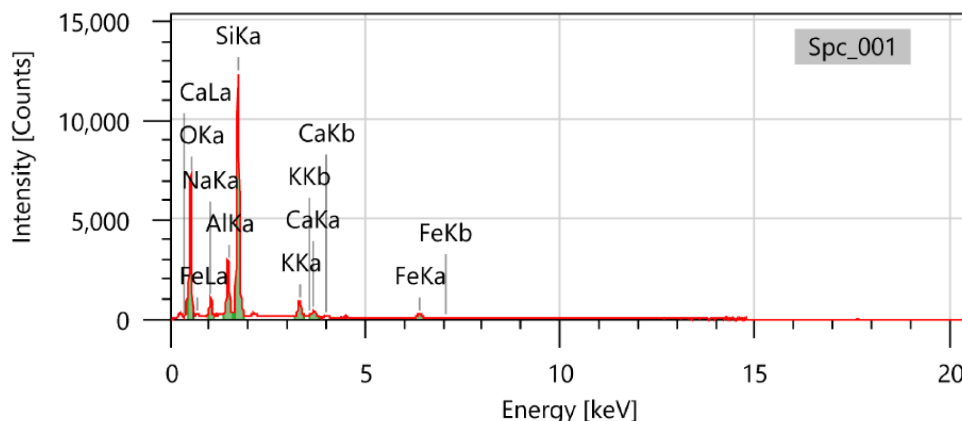


Figure 3—SEM/EDX spectrum of granite waste powder.

The SEM images at 5, 20, 50, 200, and 500 μm scales illustrate the granite waste powder's morphology at different magnifications. At 5 and 20 μm , fine grains and connected pores are clearly visible, showing a highly porous, rough texture. At 50 μm , the structure appears more aggregated with visible clusters. At 200 and 500 μm , larger-scale agglomerates and heterogeneity become evident, indicating a wide range of particle sizes and a complex, irregular surface that may enhance its performance as a bridging material in drilling fluids.

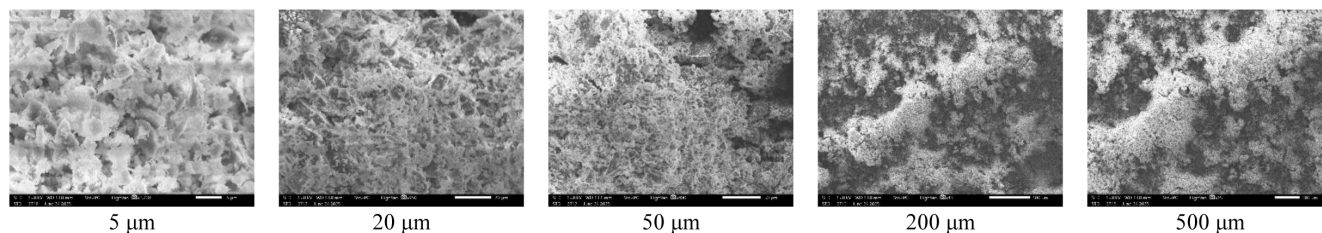


Figure 4—SEM image of granite waste powder at different magnification.

Table 1—XRF analysis of the elemental composition of granite waste

Spectrum	Concentration, wt %
Na	4.15
Al	9.35
Si	54.6
P	0.17
S	0.56
K	19.5
Ca	2.62
Ti	0.31
Cr	0.01
Fe	8.68
Ni	0.02
Cu	0.04
Zn	0.04

Drilling Fluid Preparation

Table 2 outlines the compositions of four water-based drilling fluid formulations designed to assess the performance of granite waste as a natural fluid loss additive. All mud systems (Mud-0 to Mud-3) were prepared with a consistent base formulation that includes water, bentonite, PAC-R, starch, XC polymer, potassium chloride (KCl), and barite to ensure comparable rheological and filtration behavior. Minor additives such as defoamers, soda ash, and potassium hydroxide (KOH) were used uniformly across all systems to maintain pH, reduce hardness, and stabilize the fluid structure. Mud-0 served as the baseline control with no fluid loss additive, while Mud-1 incorporated 5 ppb of conventional calcium carbonate (CaCO_3) as a reference bridging material. In Mud-2, we replaced all the barite with granite waste. In Mud-3, we used a 50% barite and 50% granite waste mixture. The granite waste used in these tests was cleaned, dried, ground, and sieved to control its particle size distribution before incorporation into the mud. This comparative formulation strategy allows for direct performance assessment of granite waste versus a conventional additive. The formulation design isolates the impact of granite waste on filtration properties, rheology, and compatibility with the fluid system, enabling systematic evaluation of its viability as a biodegradable, cost-effective alternative for fluid loss control in drilling operations.

Table 2—Composition of drilling fluid formulations with varying granite waste and CaCO_3

Material	Mud-0	Mud-1	Mud-2	Mud-3
Water	325.8	325.8	325.8	325.8
Defoamers	0.1	0.1	0.1	0.1
Soda Ash	0.5	0.5	0.5	0.5
XC	0.5	0.5	0.5	0.5
Bentonite	4	4	4	4
Starch	6	6	6	6
KOH	0.5	0.5	0.5	0.5
PAC-R	1	1	1	1
KCl	20	20	20	20
Granite waste	0	0	102.875	51.4375
CaCO_3 (25, 50 μm)	0	5	5	5
Barite	102.875	102.875	0	51.4375

Drilling Fluid Properties

For each formulation, density and pH measurements were recorded, and viscosity was measured at 120 °F using standard lab equipment. Based on the preliminary results, additional formulations may be prepared to optimize the performance of the granite waste additive. The best performing systems will be further evaluated under high-pressure high-temperature (HPHT) conditions. These HPHT tests will be carried out using a ceramic disk with 20 μm pore size at 230 °F and 300 psi to simulate real-world downhole environments.

Results and Discussion

This section presents and discusses the results obtained from the laboratory evaluation of granite waste as a natural fluid loss additive in water-based drilling fluids. The key properties measured include density, pH, rheological behavior, and filtration performance under high-pressure high-temperature (HPHT) conditions. The results from formulations containing granite waste are compared to those using conventional calcium

carbonate as a reference additive. Emphasis is placed on how the inclusion of granite waste at different concentrations affects the fluid's performance, particularly in terms of fluid loss control and filter cake quality. The observations are analyzed to assess the suitability of granite waste as an environmentally friendly and cost-effective alternative to traditional lost circulation materials.

Density and pH

Table 3 summarizes the measured density and pH values for the four mud formulations. As the granite waste content decreased from Mud 2 to Mud 3, density increased from 10.6 ppg to 11.0 ppg, while pH slightly decreased from 10.1 to 10.0. This density increase is due to the partial replacement of barite with granite waste, which changes the overall solids composition. The slight reduction in pH indicates a lower buffering capacity of granite waste compared to traditional alkaline additives. Although the pH remains within acceptable limits for water-based muds, further adjustments may be required in formulations with higher waste content to maintain chemical stability and performance.

Table 3—Measured density and pH values for different mud formulations

Formulation	Density (ppg)	pH
Mud -0	10.4	11.1
Mud -1	10.4	11.2
Mud -2	10.6	10.1
Mud -3	11.0	10.0

Filtration Behavior and Filter Cake Properties

Filtration Behavior for Different Weighting Materials. The HPHT (High-Pressure High-Temperature) filtration tests evaluated the filtration behavior of four drilling fluids: Mud-0, Mud-1, Mud-2, and Mud-3. Each mud was tested under identical conditions (Temperature: 230°F, Differential Pressure: 300 psi, Ceramic Disk: 20 Micron) for a duration of 30 minutes. The fluid loss measurements, depicted in Fig 5, clearly demonstrated variations in filtration performance:

- Mud-0 exhibited the highest fluid loss volume, reaching 8.70 mL.
- Mud-1 demonstrated moderate fluid loss, with a final volume of 7.10 mL.
- Mud-2 achieved the lowest fluid loss among all tested samples, with a final volume of 5.00 mL.
- Mud-3 showed improved performance with a lower fluid loss volume of 6.80 mL.

These results confirm the significant influence of mud composition on filtration behavior, highlighting Mud-2 as the most effective in minimizing fluid loss under HPHT conditions.

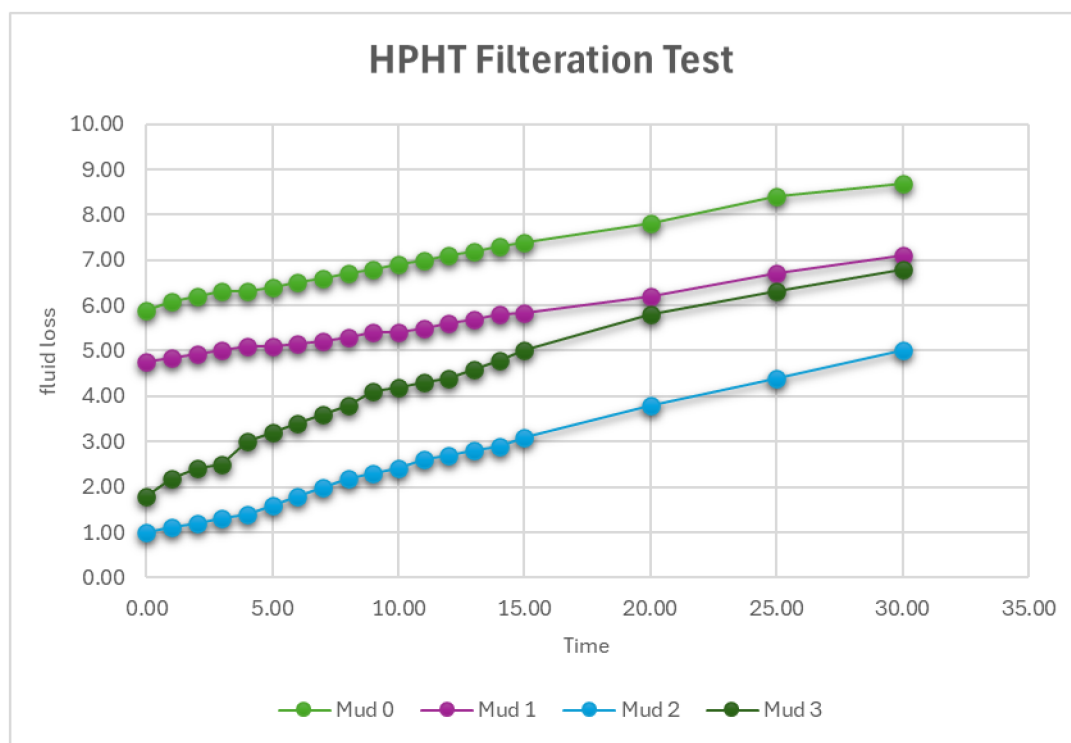


Figure 5—HPHT Filtration Test.

Analysis of HPHT Filtration Chart. The accompanying HPHT filtration chart visually represents the filtration dynamics of Mud-0, Mud-1, Mud-2, and Mud-3 over a 30-minute period. Initially, all muds displayed a rapid fluid loss that gradually stabilized. Mud-0 consistently experienced the highest fluid loss rate, whereas Mud-2 showed the lowest fluid loss rate throughout the testing period, underscoring the impact of mud composition on fluid retention capabilities.

Comparison of Filter Cake Thickness Between Mud 2 and Mud 3. The filter cake properties also indicated notable improvements in Mud-2 compared to Mud-3, with an enhancement of approximately 40%. Supporting data included:

- Initial ceramic thickness was identical (6.20 mm) for mud samples.
- Post-filtration measurements showed combined ceramic and filter cake thicknesses of 2.78 mm for Mud-2 and 3.18 mm for Mud-3.
- Notably, the calculated dry filter cake weight was considerably lower for Mud-2 (0.75g) versus Mud-3 (2.07g), signifying a thinner, more efficient filter cake developed by Mud-2.

This data, alongside the graphical representation, underscores the improved filtration management capability of the weighting material used in Mud-2, as evidenced by both quantitative measurements and visual analysis.

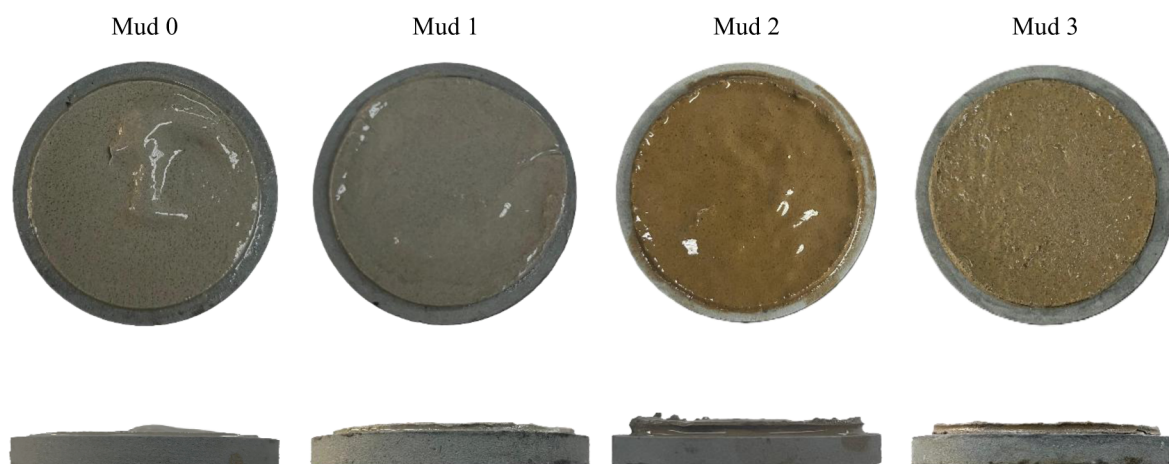


Figure 6—Filter cakes from HPHT test showing side and top views of Mud 0 and Mud 3.

Rheology and Viscosity

Plastic Viscosity and Yield Point. Fig. 7 presents a comparison of the plastic viscosity (PV) and yield point (YP) values for four mud formulations: Mud-0 (control), Mud-1 (100% barite), Mud-2 (50% barite + 50% granite waste), and Mud-3 (100% granite waste). PV increases progressively from Mud-0 through Mud-3, with the lowest values observed in the control sample (Mud-0) and the highest values recorded in the formulation containing 100% granite waste (Mud-3). Specifically, the PV rises from 10 cP in Mud-0 to 14 cP in Mud-3, while the YP increases from 17 lb/100 ft² to 21 lb/100 ft² across the same sequence. The intermediate formulations (Mud-1 and Mud-2) show gradual increases in both parameters, reflecting the influence of bridging material composition on the rheological properties of the drilling fluids. These results indicate that the incorporation of granite waste either partially or fully replacing barite not only maintains but enhances the rheological properties of the mud system in term of YP. The observed partial increases in YP and while maintaining the PV with the range suggest improved carrying capacity, which are desirable characteristics for wellbore cleaning and suspension of drill cuttings during drilling operations.

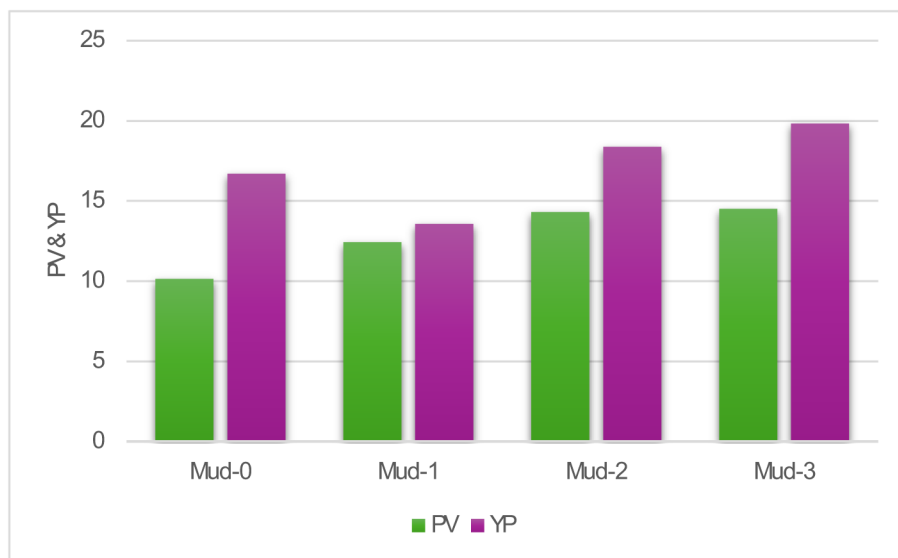


Figure 7—Plastic viscosity and yield point comparison among different mud formulations.

Gel Strength. Gel strength is crucial for maintaining fluid stability during periods when drilling activity is paused. Fig 8 illustrates the evolution of gel strength over time at 10 seconds, 10 minutes, and 30 minutes for

four mud formulations: Mud-0 (control), Mud-1 (100% barite), Mud-2 (100% granite waste), and Mud-3 (50% barite + 50% granite waste).

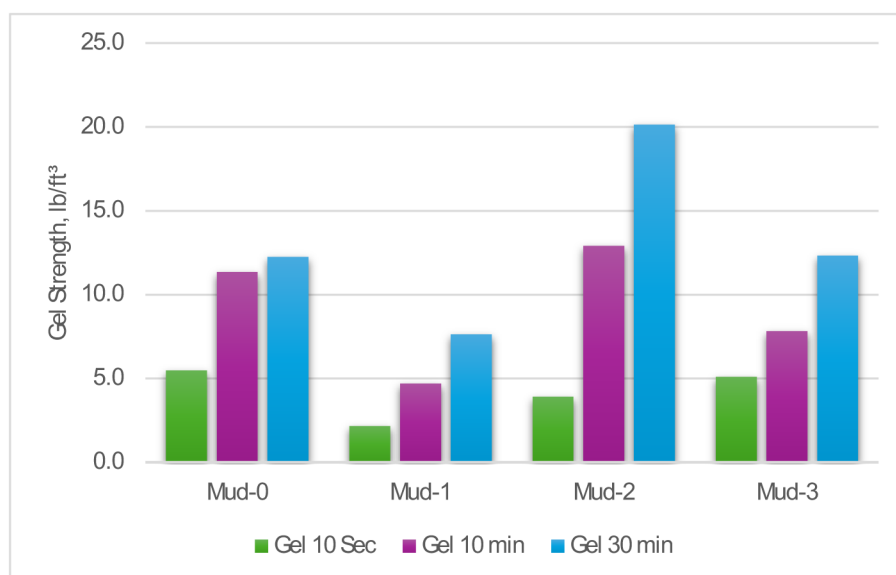


Figure 8—Gel strength measures at 10 seconds, 10 minutes, and 30 minutes for various mud formulations.

The results reveal notable variations in gel strength behavior depending on the bridging material composition. Mud-0 (control) exhibits the highest initial (10 sec) and 10-minute gel strengths, as well as substantial gel strength at 30 minutes, indicating strong structure-building capacity in the absence of added bridging agents. In contrast, Mud-1 (100% barite) displays the lowest gel strengths across all time intervals, suggesting reduced structure development and thixotropy when barite is used exclusively.

Mud-2, formulated with 100% granite waste replacing barite, shows a marked increase in gel strength over time, reaching the highest value at 30 minutes (~21 lb/ft³). This indicates improved structural stability and better ability to suspend solids over long static periods, which is valuable for wellbore cleaning and cuttings suspension during drilling interruptions.

Mud-3, containing a 50/50 blend of barite and granite waste, achieves intermediate gel strengths at all measured intervals. It performs better than the barite-only system but does not match the peak values seen in Mud-2. These results suggest that higher granite waste content can significantly enhance the gel structure development of drilling fluids, optimizing both short-term and long-term suspension properties.

Conclusions

This study demonstrated the technical feasibility and environmental advantages of incorporating granite waste powder as a biodegradable fluid loss control additive in water-based drilling fluids. Through comprehensive laboratory evaluations, the performance of granite waste-based formulations was benchmarked against conventional additives such as calcium carbonate. Key findings indicate that granite waste powder, particularly at higher concentrations, effectively reduces fluid loss while maintaining desirable rheological properties within operational limits. Compared to the conventional formulation (Mud-1), the granite waste-based mud (Mud-2) exhibited comparable filtrate volumes but produced thinner, more efficient filter cakes. This is particularly advantageous for minimizing formation damage and improving sealing efficiency. HPHT filtration results reinforced these observations, highlighting the potential of granite waste derivatives under downhole conditions. Rheological assessments revealed that Mud-3 maintained moderate plastic viscosity and yield point values, reducing internal friction while ensuring adequate cuttings transport. Moreover, gel strength results indicated improved stability with

minimal thickening over time, facilitating easier reactivation after drilling pauses. Overall, granite waste powder proved to be a cost-effective, sustainable additive with strong potential to replace traditional materials in drilling fluid formulations. Its use supports circular economy practices by valorizing agricultural waste and reducing the environmental impact of drilling operations. Future research may focus on long-term microbial stability, field-scale validation, and chemical compatibility across diverse reservoir conditions to advance commercial adoption.

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